



Project Report On “SMART VOTING SYSTEM USING PYTHON AND OPEN CV”



SUBMITTED BY

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Under the guidance of

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ACKNOWLEDGEMENT

Summer training (04th June 2019 – 01th July 2019) at DLW was a learning experience for me. It was a privilege to learn in such encouraging atmosphere with the people dedicated to work, working in tight schedule and dealing with latest technology. The success and outcome of this project required a lot of guidance and assistance from many people and I am privileged to have this all along the completion of my project. All that I have done is only due to such supervision and assistance and I would not forgot to thank them.

I respect and thank **DLW, VARANSI** for providing me an opportunity to do the project and giving us all support and guidance, which made me, complete the project duly.

I am extremely thankful to our project guide **Er. Kamalesh Kumar Verma**, who took keen interest on our project work and guided us all along, until the completion for developing a good system, although he had busy schedule managing the corporate affairs.

I am thankful to and fortunate enough to get constant encouragement, support and guidance from all colleagues who helped in successfully completing our project work. I would like to extend our sincere esteems to all the staff of DLW for their timely support.

CERTIFICATE

This is to certify that **Anurag Srivastava** of **B. Tech Information Technology (6th Semester)** branch from **Sikkim Manipal Institute of Technology, Majitar** has successfully completed **4 weeks** of industrial training from **04th June 2019** to **01th July 2019**. During this period, he successfully completed project on the topic **SMART VOTING USING USING PYTHON AND OPENCV** under the guidance of “**Er. Kamlesh Kumar Verma**”.

Project Guide Signature

ABOUT THE ORGAINSATION

The Diesel Locomotive Works (DLW) in Varanasi, India, is a production unit owned by Indian Railways. It manufactures diesel- electric locomotives and its spare parts. It is located on DLW-BHU road of the metropolitan city of Varanasi.

Website	: <i>www.diesellocomotive.com</i>
Annual production	: 125 Locomotive
Total number of Staff	: 5947
Workshop land area	: 89 hectares
Electrical power requirement	: 3468 KVA
Electrical energy consumption	: 19.8 Million

Company:

Established in 1961. It rolled out its first locomotive on 3 January 1964, after three years. The name of its first locomotive was KUNDAN. It manufacturers locomotives which are variant based on the original ALCO designs dating to 1960s and the GM EMD designs of the 1990s. DLW has an annual production capacity of 250 locomotives and plans to increase it to 275 based on the current demand. Since inception, DLW has produced total 8099 locomotives (up to 31 January 2018) of various types. DLW now manufactures “More than one locomotive a day”. DLW manufactures 334 diesel- electric locomotives in last financial year 2016-17, which is the highest locomotive production by any loco manufacturing unit in India and bagged the “Best Production Unit Shield” for the second successive year.

Products:

DLW locomotives have power outputs ranging from 2,600 horsepower (1,900 kW) to 5,500 horsepower (4,100 kW). Currently DLW is producing EMD GT46MAC and EMD GT46PAC locomotives under license from Electro- Motive Diesels (formerly GM-EMD) for Indian Railways. Some of its EMD locomotives products are WDP4D, WDP4, WDG4D, WDG5 and others as of June 2015. DLW recently started producing HEP capable WAP-7 high horsepower Electric Locomotives. It will also produce WDG4G (ES43ACmi, Customized for IR) Locomotives from General Transportation.

Market:

Besides the Indian Railways, it regularly exports diesel-electric locomotives and has supplied locomotives to other countries such as Sri Lanka, Nepal, Bangladesh, Mali, Senegal, Sudan, Tanzania, Angola, and Vietnam and also to a few users within India, such as port, large power and steel plants and private railways.

Ancillary:

In July 2006, DLW outsourced manufacture of some passenger and freight locomotives to Parel Workshop, Central Railway, Mumbai.

The **Diesel Locomotive Works**, a production unit of Indian Railways converted the diesel loco into electric loco with indigenous technology under “Make in India” initiative. The work on this project started on December 22, 2017 and the locomotive was successfully rolled out on February 28, 2018.

Parts of rail engine are made here and get distributed all over the country. Previously, they only manufacture diesel engine but now they also manufacture electric engine.

Some Famous Engine Build by the Organisation:

- **KUNDAN:** First successful loco engine made by DLW.
- **BHARTIYA:** One of the heaviest engine. Its number is 6073 YDM.

About The Technical Training Centre:

DLW organisation also provides summer vocational training to engineering and polytechnic. Based on the branch of the students they allot the particular department.

Some of the department, where they provide training:

- IT centre
- Machine Shop
- Telephone Exchange
- Electronic lab
- SCADA

ABSTRACT

The motto of the project is to build an enterprise site, which should be able to allow people to cast their vote online. The project consists of two parts:

Admin Login and Voter Login.

Administrator can perform tasks such as adding the positions for which representatives are to be elected, adding the candidates for various positions and adding the voters who are eligible to vote.

The voters need to collect their login credentials from the administrator, which will be required for logging in into the system. Once the user gets the login credentials, the user has to add its face in the database, which allows user to login using their face along with the traditional passwords. This feature adds more security and is an efficient way to login. In addition, the Smart Voting System allows the user to vote from anywhere with the same authentication as we have in the offline system due to face recognition.

It gives authority to the admin, which manages the database and keep record of all the entries and people who will vote but the admin cannot vote unless he is a voter too or bypass the login page since it requires face to login. The database consist of the voter's information along with the face dataset, which it verifies during the time of giving vote. This system makes it easier for user to cast vote, also it accessible from anywhere and everywhere, and is more secure and liable.

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CHAPTER-1

INTRODUCTION

1.1. INTRODUCTION:

The project is developed with the intent of simplifying the process of election for various positions in any enterprise. Any employee of the organization can vote for the person he wants to for the available positions.

1.2. OBJECTIVE:

- Reviewing the existing/current voting process or approach used for polling in any organization.
- Coming up with an automated voting system.
- Use facial recognition as dual authentication.
- Implementing an automated/online voting system;
- Validating the system to ensure that only registered voters are allowed to vote.

1.3. SCOPE:

This product has great future scope. This project also provides security with the use of Login-id and Password, so that any unauthorized users cannot use your account. Only authorized users who have proper access authority can access the website.

1.4. BENEFITS:

- The admin can fully control the election process without any difficulty or work overload.
- It saves time by eliminating the amount of manual work required.
- Less prone to errors.
- Information available on time.
- Reduced work force.

1.5. TARGET AUDIENCE:

Any enterprise which needs to conduct election or polling for various positions in the organization.

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PROJECT ANALYSIS

2.1. APPLICATION REQUIREMENT:

- **Operating System:-** Linux (Ubuntu)
- **Database :-** SQL
- **Front End :-** HTML, CSS, JavaScript
- **Backend :-** Python, Open CV

2.2. HARDWARE REQUIREMENT:

- **Main Memory :-** Minimum 1GB
- **CPU :-** min Intel i3 based
- **Hard Disk Space :-** min 40GB
- **Web Cam**
- **Monitor :-** any standard display
- **Keyboard :-** any standard keyboard
- **Mouse :-** any standard mouse

CHAPTER-3

FEASIBILITY REPORT

Feasibility is an important phase in the software development process it enables the developers to have an assessment of the product being developed It refers to the feasibility study of the product in terms of outcomes of the product, operational required for implement ting it. Feasibility study should be performed based on various criteria and parameters. The various feasibility studies are:

- **Economic Feasibility**
- **Operational Feasibility**
- **Technical Feasibility**

3.1. Economic Feasibility:

It refers to the benefits or outcomes we are deriving from the product as compared to the total cost we are spending for developing the benefits are more or less the same as the older system then it is not feasible to develop the product.

The product is economical feasible:

- **Reduces the processing time**
- **Reduces the work load**
- **Administrative will be effective**

3.2. Operational Feasibility:

It refers to the feasibility of the product to be operational. Some products may work very well at the design and implementation but many fail in the real time environment. It introduces the study of human resources required and their technical expertise. This provides consistent and integrated data management. It also provides information at all levels of people.

3.3. Technical Feasibility:

The system is self-explanatory and does not need any sophisticated training. A system has been built by concentrating on the graphical user interface concepts, the application can also be handled very easily with a novice uses. The overall time that a user needs to get trained is less than 15 minutes.

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SOFTWARE REQUIREMENT SPECIFICATION

4.1. REQUIREMENT SPECIFICATION:

System requirements are expressed in a software requirement document. The Software requirement specification (SRS) is the official statement of what is required of the system developers. This requirement document includes the requirements definition and the requirement specification. The software requirement document is not a design document. It should set out what the system should do without specifying how to do. The requirement set out in this document is complete and consistent.

A system should meet the following requirements for it to run the web app:

- **Web browsers: Mozilla Firefox, Google chrome**
- **Database: MYSQL**
- **Python with Django**
- **Open CV**
- **Operating System: Linux or Mac**

4.2. USER REQUIREMENTS FROM THE PROPOSED SYSTEM:

The web app should:

- a) Be able to display all registered voters in the database to the SYSTEM ADMIN(s) as per their access rights and privileges.
- b) Have a user-friendly interface and user guides understandable by people of average computer skills.
- c) Be robust enough so that users do not corrupt it in the event of voting.
- d) Be able to handle multiple users at the same time and with the same efficiency, this will cater for the large and ever growing population of voters.

4.3. FUNCTIONAL REQUIREMENTS:

- a) Secure storage and retrieval of voters' details from the database.
- b) Enable secure login of voters, i.e. to say non- legitimate voters should not be allowed to login to the app.
- c) Maintaining and manipulating records in database through functions like add, delete, etc
- d) Validate and verify input and output data.

4.4. ERROR AVOIDANCE:

At this stage care is to be taken to ensure that input data remains accurate from the stage at which it is recorded unto the stage in which the data is accepted by the system. This can be achieved only by means of careful control each time the data is handled.

4.5. ERROR DETECTION

Even though every effort is made to avoid the occurrence of errors, still a small proportion of errors are always likely to occur these types of errors can be discovered by using validations to check the input data.

4.6. DATA VALIDATION:

Procedures are designed to detect errors in data at a lower level of detail. Data validations have been included in the system in almost every area where there is a possibility for the user to commit errors. The system will not accept invalid data. Whenever an invalid data is keyed in the system immediately prompts the user and the user has to again key in the data and the system will accept the data only if the data is correct. Validations have been included where necessary. The system is designed to be a user friendly one.

4.7. USER INTERFACE DESIGN:

It is essential to consult the system users and discuss their needs while designing the user interface. USER INTERFACE SYSTEM CAN BE BROADLY CLASSIFIED AS

- User initiated interface the user is in charge, controlling the progress of the user/computer dialogue.
- Computer initiated interfaces: In the computer initiated interfaces the computer guides the progress of the user/computer dialogue. Information is displayed and the user response of the computer takes action or displays further information.

4.8. PERFORMANCE REQUIREMENTS:

Performance is measured in terms of the output provided by the application. Requirement specification plays an important part in the analysis of a system. Only when the requirement specifications are properly given, it is possible to design a system, which will fit into required environment. It rests largely in the part of the users of the existing system to give the requirement specifications because they are the people who finally use the system. This is because the requirements have to be known during the initial stages so that the system can be designed according to those requirements.

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PROJECT DESIGN

5.1. SOFTWARE ENGINEERING PARADIGM APPLIED

5.2. Reliable System:

There are two levels of reliability. The first is meeting the right requirement. A carefully and through systems study is needed to satisfy this aspect of reliability. The second level of systems reliability involves the actual working delivered to the user. At this level, the systems reliability is interwoven with software engineering and development. There are three approaches to reliability.

1. Error avoidance: Prevents errors from occurring in software.
2. Error detection and correction: In this approach errors are recognized whenever they are encountered and correcting the error by effect of error of the system does not fail.
3. Error tolerance: In this approach errors are recognized whenever they occur, but enables the system to keep running through degraded perform or Appling values that instruct the system to continue process.

5.3. Maintenance:

The key to reducing need for maintenance, while working, if possible to do essential tasks.

- More accurately defining user requirement during system development.
- Assembling better systems documents.
- Using some effective methods for designing, processing, login and communicating information with project team members.
- Making better use of existing tools and techniques.
- Managing system engineering process effectively.

5.4. Output Design:

One of the most important factors of an information system for the user is the output the system produces. Without the quality of the output, the entire system may appear unnecessary that will make us avoid using it possibly causing it to fail. Designing the output should process the in an organized well throughout the manner. The right output must be developed while ensuring that each output element is designed so that people will find the system easy to use effectively.

The term output applying to information produced by an information system whether printed or displayed while designing the output we should identify the specific output that is needed to information requirements select a method to present the formation and create a document report or other formats that contains produced by the system.

Types of output:

Whether the output is formatted report or a simple listing of the contents of a file, a computer process will produce the output:

- A Document
- A Message
- Retrieval from a data store
- Transmission from a process or system activity
- Directly from an output sources

Layout Design:

It is an arrangement of items on the output medium. The layouts are building a mock up of the actual reports or document, as it will appear after the system is in operation. The output layout has been designated to cover information.

Input design and control:

Input specifications describe the manner in which data enter the system for processing. Input design features will ensure the reliability of the systems and produce results from accurate data or thus can be result in the production of erroneous information. The input design also determines whenever the user can interact efficiently with this system.

Objectives of input design:

Input design consists of developing specifications and produces for data preparation, the steps necessary to put transactions data into a usable form for processing and data entry, the activity of data into the computer processing. The five objectives of input design are

- Controlling the amount of input
- Avoiding delay
- Avoiding error in data
- Avoiding extra steps
- Keeping the process simple

Controlling the amount of input:

Data preparation and data entry operation depend on people, because labour costs are high, the cost of preparing and entering data is also high. Reducing data requirement expense. By reduce input requirement the speed of entire process from data capturing to processing to provide results to users.

Avoiding delay:

The processing delay resulting from data preparation or data entry operations is called bottlenecks. Avoiding bottlenecks should be one objective of input.

Avoiding errors:

Through input validation we control the errors in the input data.

Avoiding extra steps:

The designer should avoid the input design that cause extra steps in processing saving or adding a single step in large number of transactions saves a lot of processing time or takes mere time to process.

Keeping process simple:

If controls are more people may feel difficult in using the systems. The best-designed system fits the people who use it in a way that is comfortable for them.

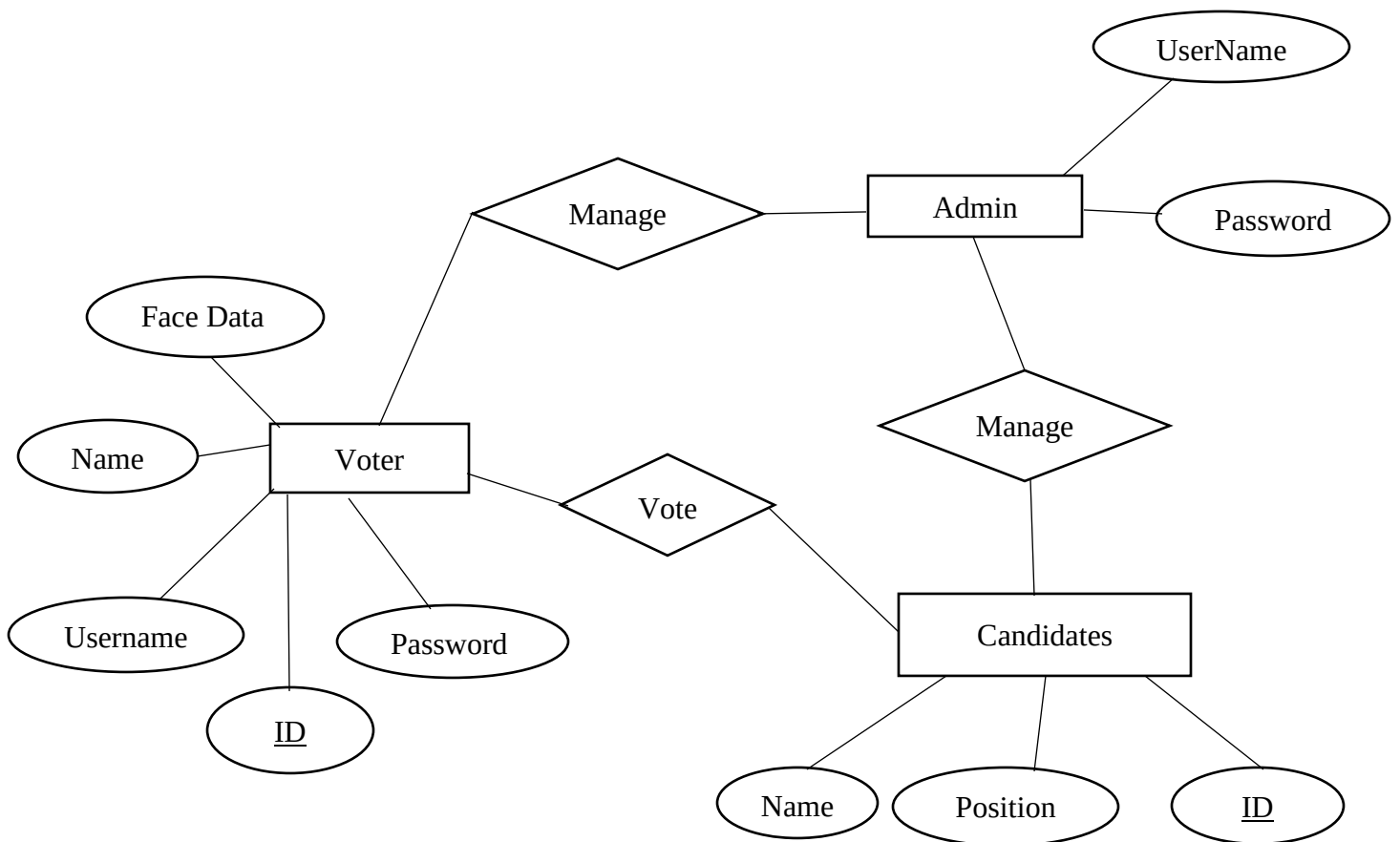
5.5. ER Diagram:

“E-R diagram are used to organize data as a relation, normalizing relations and finally obtaining a relational database model”.

ENTITY: 

RELATIONSHIPS: 

ATTRIBUTES: 

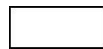


5.6. DATA FLOW DIAGRAM:

A Data Flow Diagram (DFD) is a diagram that describes the flow of data and the processes that change data throughout a system. It's a structured analysis and design tool that can be used for flowcharting in place of or in association with information. Oriented and process oriented system flowcharts. When analysts prepare the Data Flow Diagram, they specify the user needs at a level of detail that virtually determines the information flow into and out of the system and the required data resources. This network is constructed by using a set of symbols that do not imply physical implementations. The Data Flow Diagram reviews the current physical system, prepares input and output specification, specifies the implementation plan etc.

Four basic symbols are used to construct data flow diagrams. They are symbols that represent data source, data flows, and data transformations and data storage. The points at which data are transformed are represented by enclosed figures, usually circles, which are called nodes.

DATA FLOW DIAGRAM SYMBOLS:-



-Source or Destination of data



-Data Flow



-Process



-Storage

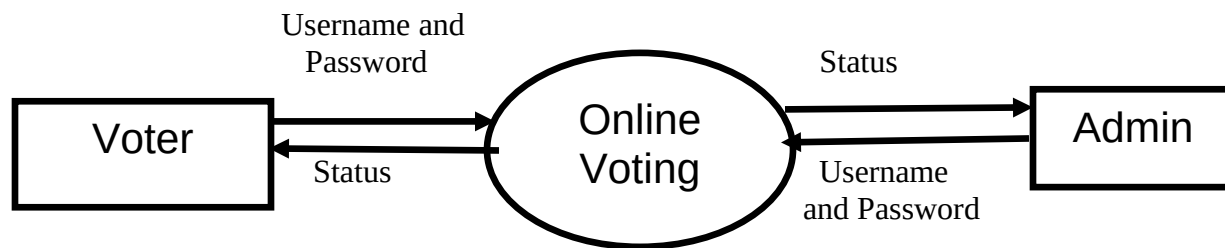
Steps to Construct Data Flow Diagrams:-

- Four steps are commonly used to construct a DFD
- Process should be named and numbered for easy reference. Each name should be representative of the process.
- The destination of flow is from top to bottom and from left to right.
- When a process is exploded in to lower level details they are numbered. o The names of data stores, sources and destinations are written in capital letters.

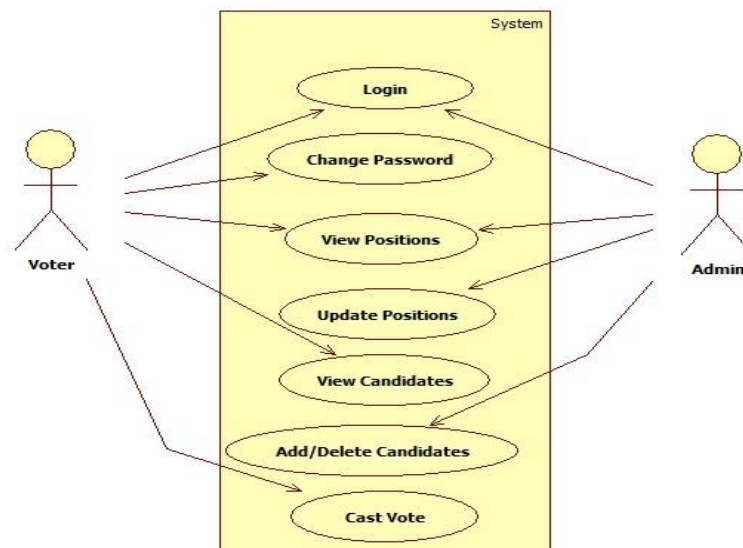
Rules for constructing a Data Flow Diagram:-

- Arrows should not cross each other.
- Squares, circles and files must bear names.
- Decomposed data flow squares and circles can have same names. Draw all data flow around the outside of the diagram.

Context Flow Diagram:



5.7. Use Case Diagram:



CHAPTER 6

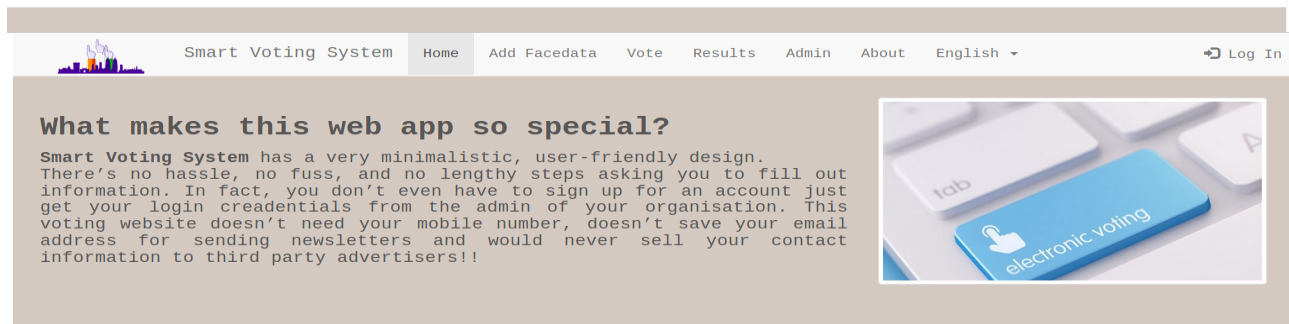
SNAPSHOTS



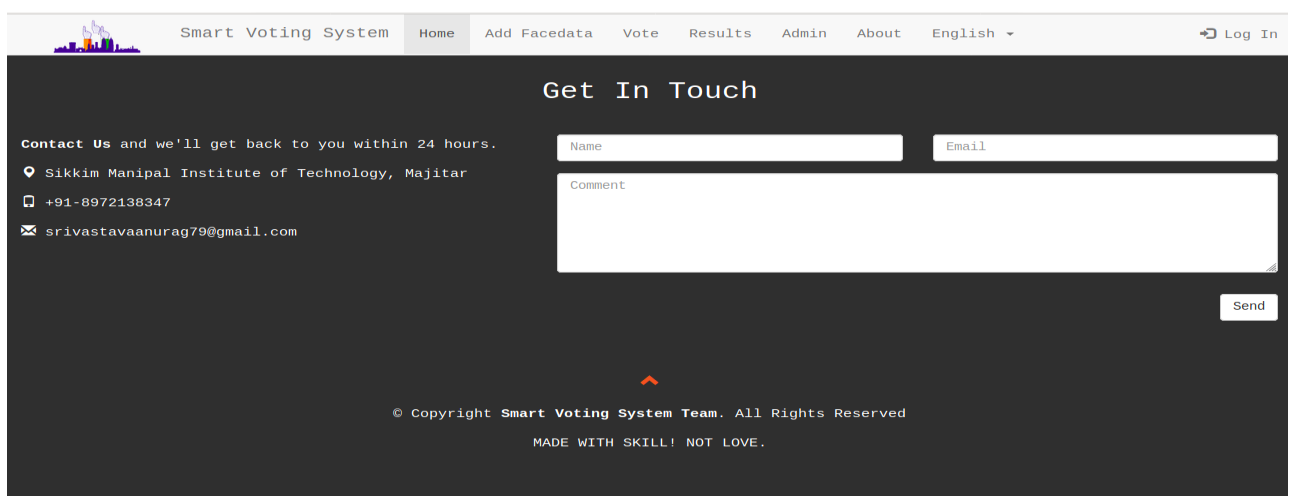
Smart voting System uses Facial Recognition along side traditional passwords which provides an additional security. This Web App is designed to serve the needs of organisation and institutions.

Smart Voting System doesn't limit the number of polls you can create. Unlike the vast majority of voting websites you can continue using the free service as often as you'd like. It's a quick and easy way to organise elections for an organisation or an institution and let your colleagues or CR's cast their vote.

Contact the Admin to get your Login Credentials. Once you get the Login Credentials you can add your Face Data and you are ready to vote.



Voting Just a **Click** Away




[Smart Voting System](#)
[Home](#)
[Vote](#)
[Results](#)
[About](#)
[English ▾](#)
[Log In](#)

LOGIN FORM

Enter your Login Credentials:

Username:
Enter Username

Password:
Enter Password

Submit Query

© Copyright Smart Voting System Team. All Rights Reserved
MADE WITH SKILL! NOT LOVE.

Smart Voting System

How to add your Facedata ?

1. Click on Create Dataset.
2. Input your Unique Id, click Submit.
3. Wait for the webcam to finish.
4. Click on Train the Classifier.

With OpenCV, LBPH

Home
 Train The Classifier
 Create Dataset


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Election Results

General Secretary

The General Secretary shall have the general responsibility for coordinating the activities of the Student Council and for directing and overseeing the publicizing of the affairs of the Student Body. She/he shall preside at all Student Council meetings. The General Secretary shall be the lead representative of the Student Body in dealings with Institute's faculty and staff, and with the community. She/he shall be the officially appointed student representative at meetings of the Board of Overseers. The General Secretary shall present an account of her/his activities to the Student Council at every Student Council meeting.



Divyanshu Gupta

Total Votes : 5

CHAPTER-7

TESTING

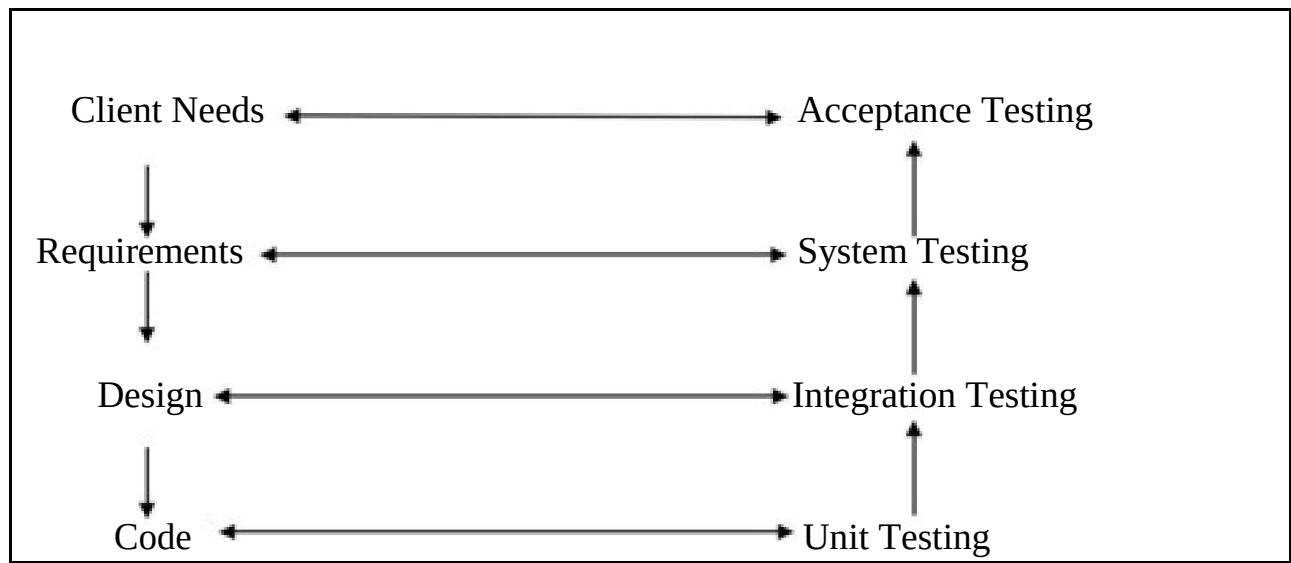
Testing is the process of executing the program with the intention of finding out errors. During testing, the program to be tested is executed with a set of test cases and the output of the programs for the test case is evaluated to determine if the program is performing as it is expected to be.

The success of testing in revealing errors in program depends critically on the test cases. In software system the use of testing is not limited to the testing phase. The results of testing are used later on during maintenance also. During testing a test suite can be used to see that modification doesn't have any undesirable effect

The basic levels of testing are:

- ✚ Unit testing
- ✚ Integration testing
- ✚ System testing
- ✚ Acceptance testing

These different levels of testing attempt to detect different types of faults. The relation of faults introduced in different phases and the different levels of testing are as shown in fig.



Levels of Testing

1. Unit Testing:

The level of testing is called unit testing. In this, different modules are tested against the specifications produced during design for the modules. Unit testing is essential for verification of the code produced during the coding phase, and hence the goal is set to test the internal logic of the modules.

2. Integration Testing:

The next level of testing is often called the integration testing. In this, many tested modules are combined into subsystems, which are then tested. The goal here is to see if the modules can be integrated properly, the emphasis being on testing interface between modules. This testing activity can be considered as testing design, and hence the emphasis on testing modules interactions.

3. System Testing:

During system testing, the system is used experimentally to ensure that the software doesn't fail, i.e. it will run according to its specifications and in the way users expect, special test data input for processing, and the results examined. A limited number of users may be allowed to use the system can see whether they try to use it in unforeseen ways.

4. Acceptance Testing:

It is sometimes performed with realistic data of the client to demonstrate that the software is working properly. Testing here focuses on the external behavior of the system. The internal logic of the program is not emphasized.

5. Validation Check:

During testing section validations checks are made. Appropriate actions are taken after testing.

CHAPTER-8

IMPLEMENTATION

Implementation is the stage of the project when the theoretical design turned into a working system. At this stage the main workload, the up heal and the major impact on the existing practices shift to user department. If the implementation stage is not carefully planned and controlled, it can cause chaos. Thus it can be considered to be the most crucial stage in achieving a new successful system and in giving the users confidence that the users confidence that the new system will work and be effective.

The implementation view of software requirements presents the real worlds manifestation of processing functions and information structures. In some cases, a physical representation is developed as the first step in software design. However, most computer-based systems are specified in a manner that dictates accommodation of certain implementation details.

Implementation involves careful planning, investigation of current system and constraints on implementation, design of methods to achieve the changeover, training of staff in the changeover procedures and evaluation of changeover methods. The first task is the implementation planning i.e. deciding the methods and time scale to adopt.

Once the planning completes, the major effort in the computer department is to ensure that the programs in the system are working properly. At the same time, the user department must concentrate on training user staff. When the staffs have been trained, a full system test can be carried out, involving both the computer and clerical procedures.

CHAPTER-9

PSEUDO CODE

```
def create_dataset(request):
    #print request.POST
    userId = request.POST['userId']
    print (cv2.__version__)
    # Detect face
    #Creating a cascade image classifier
    faceDetect =
    cv2.CascadeClassifier('/home/anurag/Downloads/SmartSearch-
    Using-Face-Recognition
    master/Facial/Facial/ml/haarcascade_frontalface_default.xml')
    #capture images from the webcam and process and detect the face
    # takes video capture id, for webcam most of the time its 0.
    cam = cv2.VideoCapture(0)

    # Our identifier
    # We will put the id here and we will store the id with a face, so that
    later we can identify whose face it is
    id = userId
    # Our dataset naming counter
    sampleNum = 0
    # Capturing the faces one by one and detect the faces and showing it
    on the window
    while(True):
        # Capturing the image
        #cam.read will return the status variable and the captured colored
        image
        ret, img = cam.read()
        #the returned img is a colored image but for the classifier to work
        we need a greyscale image
        #to convert
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
#To store the faces
#This will detect all the images in the current frame, and it will
return the coordinates of the faces
#Takes in image and some other parameter for accurate result
faces = faceDetect.detectMultiScale(gray, 1.3, 5)
#In above 'faces' variable there can be multiple faces so we have to
get each and every face and draw a rectangle around it.
for(x,y,w,h) in faces:
# Whenever the program captures the face, we will write that is a
folder
# Before capturing the face, we need to tell the script whose face it
is
# For that we will need an identifier, here we call it id
# So now we captured a face, we need to write it in a file
sampleNum = sampleNum+1
# Saving the image dataset, but only the face part, cropping the rest
cv2.imwrite('/home/anurag/Downloads/Smart-Search-Using-Face-
Recognition-
master/Facial/Facial/ml/dataset/'+str(id)+'.'+str(sampleNum)+'.jpg',
gray[y:y+h,x:x+w])
# @params the initial point of the rectangle will be x,y and
# @params end point will be x+width and y+height
# @params along with color of the rectangle
# @params thickness of the rectangle
cv2.rectangle(img,(x,y),(x+w,y+h), (0,255,0), 2)
# Before continuing to the next loop, I want to give it a little pause
# waitKey of 100 millisecond
cv2.waitKey(250)
```

CHAPTER-10

CONCLUSION

Elections in an Enterprise were earlier being carried out using ballot paper. The process was time taking and had a lot of work overload. This project helps in eliminating these problems.

This project also helps in making the election process transparent and easy. Poling for many positions can be carried out simultaneously and without any difficulty. It helps in reducing the amount of manual and paper work.

The administrator can easily **monitor** and **control** the election process. It saves time by eliminating the amount of manual work required.

Voter's location is not important but their voting is important. As data is stored in database, so data is accessible at any time. Smart voting system provides more fair and secure system and process to cast voting, irrespective of the voter's location.

Hence, this provides a perfect platform for easy login and a secure system. In addition, it allows the system to calculate the number of votes given to each candidate, making it more demanding and easier for the user.

Hence, it is evolutionary to be working with such easy to operate and less time consuming, which ignores the outer physical factors, which may affect the offline voting system. And making its availability to everyone.

CHAPTER-11

REFERENCES

- www.djangoproject.com/
- docs.djangoproject.com/en/2.2/
- www.engpaper.com/face-recognition-2018.htm
- www.w3schools.com
- www.tutorialspoint.com
- www.stackoverflow.com
- Edx- CS50